NEXUS.AI

THREAT DETECTION & SIMULATION

AI-Driven Threat Detection & Simulation Engine

Minimal cybersecurity dashboard presentation for Mentor Round 1 evaluation.

DETECT
Multi-layer threat discovery

EXPLAIN
Reasoning behind alerts

SIMULATE
Future attack scenarios

TEAM NAME	HACKATHON NAME
[Your Team Name]	**Hack Malenadu '26**

The Problem

SOC teams are drowning in alert volume without enough context.

10,000+
Alerts per day
High alert volume

60–70%
False positives
Analyst fatigue

Market Need

$200B+
Cybersecurity market

~30%
XDR CAGR

Growth pressure on modern SOC tooling

Zero-day
Blind spot for signatures
Missed novel threats

Rule-based
Legacy detection logic
Weak against APT patterns

Security teams are overwhelmed and reactive.

10k+

60–70%

Our Approach

Use case: Enterprise SOC monitoring

Focus: Mentor Round 1

DETECT

Detect

Identify threats across network, endpoint, and application layers in real time.

Cross-layer visibility

EXPLAIN

Explain

Show why an alert fired using evidence, context, and AI-assisted reasoning.

Analyst-ready evidence

SIMULATE

Simulate

Project possible next steps of an attack to support proactive defense planning.

Prevention-oriented

Core flow

Logs → AI Engine → Detection → Explanation → Simulation

Logs Network + Endpoint → **AI Engine** Normalization + features → **Detection** Threat logic → **Explanation** Context + evidence → **Simulation** Future scenarios

System Architecture

Diagram-first view of the technical pipeline

Network Logs
Packets / flows

+

Endpoint Logs
Sysmon / host events

↓

Data Normalization
Unified schema + time alignment

↓

Feature Engineering
Rates, sequences, anomalies, sessions

↓

Detection Models
Anomaly + pattern-based

→

Correlation Engine
Cross-layer aggregation

↓

Explainability + Simulation
Evidence + possible next attack steps

Pipeline Focus

- Multi-source ingestion from network and endpoint layers
- Normalized events make downstream detection consistent
- Hybrid models balance novelty detection and known patterns
- Correlation boosts confidence before escalation
- Explainability keeps analysts in control

Data Strategy

Designed for noisy, real-world security telemetry.

Core datasets

CICIDS 2017

Network intrusion behaviors for baseline and malicious flow patterns.

UNSW-NB15

Modern network attack categories for robust feature evaluation.

Synthetic endpoint logs add host-level realism.

Benign traffic

Normal enterprise activity, routine browsing, regular file access, common service communication.

Malicious traffic

Attack traces for brute force, lateral movement, exfiltration, and beacon-like sessions.

False positive scenario

Admin bulk transfer intentionally resembles exfiltration to test analyst trust and scoring behavior.

Designed to handle noisy, real-world data.

2

+ synthetic

Threat Detection Logic

Cross-layer signals increase confidence and reduce noisy escalation.

Brute force

Repeated login failures, velocity spikes, and repeated auth attempts.

Lateral movement

Unusual internal traffic, unexpected host-to-host access, privilege spread.

Data exfiltration

Large outbound transfers, atypical destinations, bursty transfer behavior.

C2 beaconing

Periodic connections, recurring low-volume callbacks, timing regularity.

Scoring & severity

Confidence	0–100
Severity	Low–High
Correlation	Cross-layer

Cross-layer correlation increases accuracy.

Execution Plan & Readiness

Focused scope for hackathon delivery.

Team roles

Data & Ingestion
Dataset prep, parsers, flow consolidation

ML & Detection
Threat logic, scoring, anomaly detection

Backend
Pipelines, APIs, event handling

Dashboard
Analyst UI, alerts, explainability views

Hackathon timeline

PHASE 1
Data + ingestion
Load datasets, create schemas, normalize logs, prepare test inputs.

PHASE 2
Detection logic
Implement rules, anomaly features, confidence scoring, and correlation.

PHASE 3
Integration + UI
Connect backend outputs to the dashboard and demo analyst workflow.

Focused, feasible, and implementable within hackathon time.